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Collapsible table

Abstract

The present disclosure relates to a collapsible table. In one or more embodiments, an apparatus includes a table. The table includes a first central segment and a second central segment. The table further includes a plurality of joints configured to facilitate movement of the first central segment and the second central segment. The movement of the first central segment and the second central segment causes one or more dimensions of the table to be modified.

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Background/Summary

INTRODUCTION

(1) The present disclosure relates to a collapsible table.

SUMMARY

(2) The present disclosure describes a collapsible table that is particularly suitable for storage and transportation in a storage compartment of a vehicle, such as front trunk (“frunk”).

(3) In one or more embodiments, an apparatus includes a table. The table includes a first central segment and a second central segment. The table further includes a plurality of joints configured to facilitate movement of the first central segment and the second central segment. The movement of the first central segment and the second central segment causes one or more dimensions of the table to be modified.

(4) In one or more embodiments, a method for deploying a table includes translating a first central segment and a second central segment of the table outwardly from one another to cause one or more dimensions of the table to be modified. The table includes a plurality of joints configured to

facilitate the translation of the first central segment and the second central segment. The method further includes placing one or more panels on the table to form a tabletop.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A illustrates a collapsible table in a collapsed configuration, in accordance with certain embodiments.
- (2) FIGS. 1B and 1C illustrate the collapsible table in a deployed configuration with various mounted accessories, in accordance with certain embodiments.
- (3) FIG. 1D illustrates the collapsible table in a partially deployed configuration, in accordance with certain embodiments.
- (4) FIG. 2A illustrates a tabletop frame of the collapsible table in the collapsed configuration, in accordance with certain embodiments.
- (5) FIG. 2B illustrates the arrangement of the legs of the collapsible table in the collapsed configuration, in accordance with certain embodiments.
- (6) FIGS. 2C and 2D illustrate the collapsible table with the legs thereof deployed and a tabletop frame thereof collapsed, in accordance with certain embodiments.
- (7) FIG. 2E illustrates the tabletop frame of the collapsible table in an intermediate configuration between the collapsed configuration and the deployed configuration.
- (8) FIG. 2F illustrates the tabletop frame of the collapsible table in the deployed configuration in accordance with certain embodiments.
- (9) FIG. 2G illustrates a panel of the collapsible table, in accordance with certain embodiments.
- (10) FIG. 3A illustrates the collapsible table positioned within a storage compartment of a vehicle while in the collapsed configuration, in accordance with certain embodiments.
- (11) FIG. 3B illustrates a collapsible table shaped to conform to a non-rectangular shape of a storage compartment of a vehicle.
- (12) FIG. 3C illustrates the collapsible table placed in a cargo area of a vehicle, in accordance with certain embodiments.
- (13) FIG. 4 is a process flow diagram of a method for deploying a collapsible table, in accordance with certain embodiments.

DETAILED DESCRIPTION

(14) Off-road vehicles enable a driver and passengers to explore exciting new locations. A vehicle may be provided with accessories, such as a roof-top or truck-bed tent, camp kitchens, and the like, to make overnight camping more comfortable. Although many accessories can be positioned in a truck bed or mounted to a roof rack, users will also need to store gear and supplies within the vehicle itself. Accordingly, it is advantageous to make gear for storage within the vehicle itself as compact as possible.

(15) The present disclosure describes a collapsible table that is well suited to be stored within a storage compartment of a vehicle, such as a rear trunk, front trunk (“frunk”), or other cargo area of a vehicle. The collapsible table may be specifically sized to be stored laid flat or oriented vertically within the storage compartment of a particular vehicle. The collapsible table includes a tabletop frame having a collapsed and deployed configuration and work surface composed of a plurality of panels that conform to the size of the tabletop frame when in the collapsed configuration. The tabletop frame and the plurality of panels may therefore be stacked on top of one another and placed in a space that is much smaller than a footprint of the collapsible table when in the deployed configuration.

(16) Referring to FIGS. 1A to 1C, a collapsible table **100** may be understood with respect to a horizontal direction **102a**, a longitudinal direction **102b**, and a vertical direction **102c** that are all

perpendicular to one another. The collapsible table **100** is described with respect to the directions **102a**, **102b**, **102c** (which may also be referred to in reference to a length, a width, and a height of the collapsible table **100**, respectively) to facilitate understanding of the relative positions and orientations of components of the table **100** with the understanding that the directions **102a**, **102b**, **102c** need not correspond to the actual orientation of the collapsible table **100** during use. In the following description, the directions **102a**, **102b**, **102c** are defined relative to a tabletop frame **104** of the table **100**.

(17) The table **100** has a collapsed configuration shown in FIG. 1A in which the tabletop frame **104** is collapsed and two, three, or more panels **106** may be stacked with the tabletop frame **104** (above or below the tabletop frame **104**) to achieve a compact size. The tabletop frame **104** has a dimension **108a** in the horizontal direction **102a** and a dimension **108b** in the longitudinal direction **102b** when in the collapsed configuration. In the illustrated embodiment, dimension **108b** is greater than the dimension **108a**, such as between 1.5 and 3 times greater.

(18) The panels **106** are generally planar members with a planar surface **106a** that, when stacked with the tabletop frame **104**, are substantially (e.g., within 5 degrees of) orthogonal to the vertical direction **102c** defined by the tabletop frame **104**. In the illustrated embodiment, the panels **106** each have a rectangular shape in a plane parallel to the planar surface **106a** thereof. The panels **106** may have edges conforming to a rectangular shape with other features extending between edges, such as one or more beveled corners (see, e.g., FIG. 3B) or rounded corners. In various embodiments, other shapes may be implemented such that each panel **106** has at least one edge lying on the rectangular shape, each panel **106** is substantially contained within the rectangular shape, and each panel **106** can abut with another panel of the plurality of panels **106**.

(19) The rectangular shape has a dimension **110a** and a dimension **110b** perpendicular to the dimension **110a** in a plane parallel to the planar surface **106a**. The dimension **110a** and the dimension **110b** may be substantially (e.g., within 5, 10, or 15 percent of) equal to the dimensions **108a**, **108b**, respectively. In some embodiments, the dimensions **110a**, **110b** are no greater than the dimensions **108a**, **108b**, respectively. In this manner, the panels **106** and tabletop frame **104** may be stacked in the compact configuration shown in FIG. 1A.

(20) Although the panels **106** are shown as having a rectangular shape, other shapes are may be implemented. For example, the panels **106** may include a rectangular panel and two semi-circular panels that together define a discorectangle in a deployed configuration. In the illustrated embodiment, the panels **106** are substantially (e.g., within 5 percent of) equal to one another.

However, other configurations may be implemented. For example, two panels with rounded or beveled edges may be used on either side of a third panel that is both smaller and rectangular.

(21) The panels **106** may be made of a material that is sufficiently strong to avoid deflection during use and withstand abrasion. The panels **106** may be made of wood, metal (e.g., aluminum or steel) or plastic (e.g., polyvinyl chloride (PVC), acrylonitrile butadiene styrene (ABS), polypropylene, acrylic, polyethylene, polystyrene, or the like). The panels **106** may be made of a composite material, such as a carbon fiber, fiberglass, Kevlar, or other composite material.

(22) The tabletop frame **104** may be made of steel, aluminum, or any other material, such as the plastic or composite materials listed above. Unless otherwise noted below, the members of the tabletop frame **104** may be made of round or rectangular tubes, such as tubes with cross-sectional dimensions of between 2 and 10 cm and a wall thickness between 2 and 4 mm for metal with plastic or composite materials possibly having greater wall thickness.

(23) Referring to FIG. 1B, while still referring to FIG. 1A, in the deployed configuration, the tabletop frame **104** has a dimension **108c** in the horizontal direction **102a** that is much larger than the dimension **108a**, such as at least two times, at least three times, or some a greater multiple. For example, as shown in FIG. 1B, the dimension **108c** is substantially (e.g., within 10 percent of) equal to an integer multiple of the dimension **108a**, where the integer multiple is equal to the number of the panels **106**, e.g., two, three, or larger value.

(24) In the deployed configuration, the panels **106** are rested on the tabletop frame **104** with the dimension **110b** thereof oriented substantially (e.g., within 5 degrees of) parallel to the longitudinal direction **102b** and the dimension **108b** of the tabletop frame **104**, with edges of adjacent panels **106** defining a gap of less than 5 mm, 2 mm, or 1 mm, or abutting one another. When resting on the tabletop frame **104**, the planar surfaces **106a** of the panels **106** may be substantially (e.g., within 2 degrees of) parallel to one another and substantially coplanar (e.g., having centers thereof within 5, 2, or 1 mm of one another along the vertical direction **102c**). The panels **106** may extend outwardly from the tabletop frame **104** in one or both of the horizontal and longitudinal directions **102a**, **102b** or may be positioned at or completely within the perimeter of the tabletop frame **104** along one or both of the horizontal and longitudinal directions **102a**, **102b**.

(25) In the deployed configuration, legs **112** may be secured to the tabletop frame **104** with the legs **112** extending downwardly from a lower surface **104b** of the tabletop frame **104** with the panels **106** resting on an upper surface **104a** of the tabletop frame **104** opposite the lower surface **104b** along the vertical direction **102c** to form a tabletop **105**. In the deployed configuration, the tabletop **105** may include a surface defined by multiple panels **106** (e.g., two panels **106**, three panels **106**, or more panels **106**), as shown in FIGS. **1B** and **1C**. In a partially deployed or collapsed configuration, the tabletop **105** may include a surface defined by a single panel **106** (e.g., with one or more panels **106** stacked underneath the single panel **106**, as shown in FIG. **1A**). The legs **112** may extend downwardly from the lower surface **104b** with a longest dimension thereof substantially (e.g., within 5 degrees of) parallel to the vertical direction **102c**. In the collapsed configuration, the legs **112** may be either (a) detached from the tabletop frame **104** or (b) pivoted such that the longest dimensions thereof is substantially perpendicular to the vertical direction **102c**, such as at an angle of between 80 and 90 degrees or between 85 and 90 degrees relative to the vertical direction **102c**. An example embodiment including legs **112** pivotally secured to the tabletop frame **104** is described in further detail below. In still other embodiments, the legs **112** are at some other non-perpendicular angle when in the deployed configuration, e.g., scissor legs.

(26) In some embodiments, one or more braces **114** may be mounted to pairs of legs **112** to resist folding of the legs **112** relative to the tabletop frame **104**. For example, in the illustrated embodiment, a brace **114** connects pairs of legs **112** that are offset from one another along the longitudinal direction **102b** and substantially aligned (e.g., within 2 cm) along the horizontal direction **102a**. Stated differently, legs **112** positioned adjacent corners that are connected by a common edge of the tabletop frame **104** may be connected by a brace **114**. The longest dimension of each brace **114** may be oriented substantially (e.g., within 5 degrees of) parallel to one of the directions **102a**, **102b** or may be positioned diagonally between legs **112** offset from one another along both of the horizontal and longitudinal directions **102a**, **102b**. The brace **114** may secure to legs **112** by threading into threaded holes formed on the legs **112** or engaging a mounting bracket or some other mounting structure secured to the legs **112**.

(27) In the illustrated embodiment, the tabletop frame **104** includes a first end segment **116a** and a second end segment **116b**. The first and second end segments **116a**, **116b** may have fixed dimensions in the longitudinal direction **102b** that does not change between the collapsed and deployed configurations. The distance between the first and second end segments **116a**, **116b** is expanded when transitioning from the collapsed configuration to the deployed configuration.

(28) One or more fasteners **118** may be used to secure the first and second end segments **116a**, **116b** to one another in the collapsed configuration. The one or more fasteners **118** may be disengaged to transition the tabletop frame **104** to the deployed configuration. For example, the illustrated fastener **118** may be used alone or with a second fastener (not shown) on an opposite side of the tabletop frame **104** along the longitudinal direction **102b**.

(29) In the illustrated embodiment, the fastener **118** may include a pivoting hook **118a** secured to the first end segment **116a** that engages a post **118b** secured to the second end segment **116b** (or vice versa). The fastener **118** may also be embodied as an over-center latch secured to the first end

segment **116a** engaging a hook secured to the second end segment **116b** (or vice versa) or a strap secured to the first end segment **116a** and engaging a buckle on the second end segment **116b** (or vice versa). The fastener **118** may be embodied as a clip portion secured to the first end segment **116a** and engaging a clip portion secured to the second end segment **116b**, such as a portion of a side-release buckle or other type of clip. The fastener **118** may also be implemented as other types of fasteners.

(30) As shown in FIGS. **1B** and **1C**, various accessories **120a**, **120b**, **120c** may be used with the table **100** when in the deployed configuration. For example, a panel **106** may define an opening **122** that receives a basin **120a**, such as the illustrated collapsible basin. The basin **120a** may either be removed from the opening **122** when the table **100** is in the collapsed configuration or the basin **120a** may remain in the opening **122** in a collapsed state when the table **100** is in the collapsed configuration. Accessories used with the table **100** may include a stove **120b** supported on the table **100**. The stove **120b** may have a length equal to or greater than the dimension **108c** of the table **100** in the deployed configuration. The stove **120b** may be a gas or electric stove, such as an electric stove powered by the battery of an electric vehicle. In some embodiments, a storage structure **120c** (FIG. **1C**) rests on or mounts to one or more of the panels **106**. The storage structure **120c** may define one or more shelves, drawers, bins, or other storage compartments for storing kitchen utensils, tools, or other items. One or more panels may define one or more registration features (e.g., registration holes, pins, mounting brackets, etc.) for mounting the illustrated accessories **120b**, **120c** or other accessories.

(31) Referring to FIG. **1D**, in some embodiments, the table **100** may be used in a partially deployed configuration in which the tabletop frame **104** remains in the collapsed configuration, i.e., having dimensions **108a**, **108b**, with the legs **112** in their deployed configuration. A single panel **106** may then be rested on the tabletop frame **104**, or multiple panels **106** stacked on top of one another.

(32) FIGS. **2A** to **2F** illustrate the process of transitioning the table **100** from the collapsed configuration to the deployed configuration as well as an example embodiment of structures of the tabletop frame **104** facilitating the transition. The process of transitioning the table **100** from the deployed configuration to the collapsed configuration is the reverse of the process of transitioning the table **100** from the collapsed configuration to the deployed configuration.

(33) Referring specifically to FIG. **2A**, the first and second end segments **116a**, **116b** define inwardly facing surfaces **202a**, **202b**. The inwardly facing surfaces **202a**, **202b** face inwardly into the tabletop frame **104** and define a space in which a plurality of links are positioned in the collapsed configuration. The inwardly facing surfaces **202a**, **202b** may be substantially (e.g., within 5 degrees of) parallel to the vertical direction **102c** or may be rounded or have some other orientation.

(34) The first end segment **116a** and the second end segment **116b** are joined to one another by collapsible sides **204a**, **204b**, respectively, that are positioned between the inwardly facing surfaces **202a**, **202b** along the horizontal direction when the tabletop frame **104** is in the collapsed configuration. The collapsible sides **204a**, **204b** may have a mirrored configuration with respect to one another. Each collapsible side **204a**, **204b** includes a plurality of joints facilitating movement of the first end segment **116a** relative to the second end segment **116b** to modify the dimensions of the tabletop frame **104**, such as from having the dimension **108a** to having the dimension **108c**. For example, each collapsible side **204a**, **204b** may include a pivotable portion **206a** that is pivotally connected by pivot **208a** to the first end segment **116a** and a pivotable portion **206b** that is pivotally connected by pivot **208b** to the second end segment **116b**.

(35) In the illustrated embodiment, the first and segments **116a**, **116b** may each include extensions **210a**, **210b**, respectively, that are rigidly secured to either end of the first and second end segments **116a**, **116b** along the longitudinal direction **102b**. The extensions **210a**, **210b** may secure to the first and second end segments **116a**, **116b** by either being monolithically formed as part of the first and second end segments **116a**, **116b** or being fastened to the first and second end segments **116a**, **116b**

by bolts, welds, or other type of fastener. The extensions **210a**, **210b** extend outwardly relative to the inwardly facing surfaces **202a**, **202b** and inwardly into the tabletop frame **104** along the horizontal direction **102a**. The pivots **208a**, **208b** may pivotally mount the pivotable portions **206a**, **206b** to the extensions **210a**, **210b**, respectively, such that the pivots **208a**, **208b** are offset inwardly from the inwardly facing surfaces **202a**, **202b**, such as by between 10 and 50 percent of the dimension **108a**.

(36) In the illustrated embodiment, outward facing (e.g., facing outwardly in the horizontal direction **102a**) surfaces of the first end segment **116a** and the second end segment **116b** are parallel to the longitudinal direction **102b** and outward facing (e.g., facing outwardly in the longitudinal direction **102b**) surfaces of the extensions **210a**, **210b** are parallel to the horizontal direction **102a**. However, other configurations may also be used, including extensions **210a**, and/or extensions **210b** with outward facing surfaces that define a diagonal with respect to the outward facing surfaces of the first end segment **116a** and the second end segment **116b**, such as for use in the embodiment shown in FIG. 3B.

(37) In the illustrated embodiments, the extensions **210a**, **210b** define inwardly facing cavities **212** into which the pivotable portions **206a**, **206b**, respectively, are inserted. For example, the extensions **210a**, **210b** may be implemented using portions of U-channel bars with the opening of the U-channel facing inwardly into the tabletop frame **104**. The pivots **208a**, **208b** may be implemented as pins, bolts, or other fasteners passing into the cavities **212** and through the pivotable portions **206a**, **206b**, respectively.

(38) The plurality of joints of each collapsible side **204a**, **204b** may further include a central segment **214**. The pivotable portions **206a**, **206b** of each collapsible side **204a**, **204b** extend from the pivots **208a**, **208b**, respectively to the central segment **214** and are pivotally secured to the central segment **214** by pivots **216a**, **216b**, respectively. In the illustrated embodiment, the central segment **214** defines a cavity **218** that receives the pivotable portions **206a**, **206b**. For example, the central segment **214** may be implemented as a U-channel bar with the opening of the U-channel facing outwardly from the tabletop frame **104**, i.e., the opposite direction from the cavities **212** of the extensions **210a**, **210b**. The pivots **216a**, **216b** may be implemented as pins, bolts, or other fasteners passing into the cavity **218** and through the pivotable portions **206a**, **206b**, respectively.

(39) The offset along the horizontal direction **102a** between the pivots **208a**, **208b** and the first and second end segments **116a**, **116b**, the offset along the horizontal direction **102b** between the pivots **216a**, **216b** securing the pivotable portions **206a**, **206b** to the central segment **214**, and the distance between the pivots **208a**, **208b** and the pivots **216a**, **216b** along each pivotable portion **206a**, **206b** defines the dimension **108c** of the table **100** when in the deployed configuration. The extent of the extensions **210a**, **210b** outwardly from the first and second end segments **116a**, **116b** in the horizontal direction **102a** may define the dimension **108a** of the tabletop frame **104** in the collapsed configuration.

(40) The offset between the inwardly facing surfaces **202a**, **202b** along the horizontal direction **102a** in the collapsed configuration may define the limit to the length of the central segments **214**. For example, the central segments **214** may have lengths of between 50 and 98 percent of the dimension **108a** in some embodiments, though other configurations may be implemented. The dimension **108b** may define the limit to the length of the pivotable portions **206a**, **206b**, i.e., the length at which the central segments **214** of the collapsible sides **204a**, **204b** will interfere with one another, though overlap is permitted in other embodiments. The offset between the pivots **216a**, **216b** on the central segments **214** facilitates increasing the dimension **108c** without the need to increase the dimension **108b** to accommodate longer pivotable portions **206a**, **206b**. Stated more generally, the length of the extensions **210a**, **210b** and length of the central segments **214** between the pivots **216a**, **216b** are variables that can be adjusted to achieve desired dimensions **108a** and **108c** for the tabletop frame **104** for a given dimension **108b**.

(41) Control arms **220a**, **220b** may pivotally mount to the central segment **214** and to the

extensions **210a**, **210b**, respectively, of each collapsible side **204a**, **204b**. The control arms **220a**, **220b** constrain the central segment **214** to move substantially (e.g., within 10 degrees of) parallel to the horizontal direction **102a**. This constraint enables a user to transition the tabletop frame **104** more easily between the deployed and collapsed configuration by eliminating some degrees of freedom and therefore eliminating potentially uncontrolled or unpredictable pivoting of members of the tabletop frame **104**.

(42) In the illustrated embodiment, each control arm **220a**, **220b** secures to a portion of the central segment **214** extending outwardly from the pivots **216a**, **216b** by means of pivots **222a**, **222b**, respectively. The control arms **220a**, **220b** are also pivotably secured to the extensions **210a**, **210b** by means of pivots **224a**, **224b**, respectively. In the illustrated embodiment, the control arms **220a**, **220b** are implemented as flat plates that are thinner in the vertical direction **102c** than in the horizontal and longitudinal directions **102a**, **102b**. The control arms **220a**, **220b** may be straight as in the illustrated embodiment or may be curved in a plane parallel to the horizontal and longitudinal directions **102a**, **102b** to provide clearance for the pivots **208a**, **208b** when the collapsible sides **204a**, **204b** are in the deployed configuration. In the illustrated embodiment, the control arms **220a**, **220b** are mounted on lower surfaces of the central segment **214** and extensions **210a**, **210b** facing away from panels **106** resting on the tabletop frame **104**, though other arrangements are also possible.

(43) Referring to FIG. 2B, while still referring to FIG. 2A, a pair of legs **112** may be pivotably mounted to each of the first and second end segments **116a**, **116b**. FIG. 2B is shown and described with respect to the first end segment **116a** with the second end segment **116b** being configured in the same manner. A leg **112a** of a pair of legs **112** is mounted by pivot **226a** to the first end segment **116a** and the leg **112b** of the pair of legs **112** is mounted to the first end segment **116a** by pivot **226b**. The axis of rotation of the pivots **226a**, **226b** may be substantially (e.g., within 5 degrees of) parallel to the longitudinal direction **102b**. The pivots **226a**, **226b** are offset from one another along the longitudinal direction **102b** by an amount less than the dimension **108b**, such as between 75 and 95 percent of the dimension **108b**. The pivots **226a**, **226b** may also be offset from one another in the vertical direction **102c**, such as by an amount approximately (e.g., within 5 percent of) equal to the thickness of the legs **112a**, **112b** in the vertical direction when oriented with the long dimensions thereof substantially (e.g., within 5 degrees of) parallel to the longitudinal direction **102b**. In this manner, the legs **112a**, **112b** may be placed substantially parallel to the longitudinal direction **102b**. The leg **112b** mounted to the pivot **226b** that is lower when the table **100** is oriented with the legs **112** in the deployed configuration and resting on a support surface may be shorter than the other leg **112a** of the pair of legs **112** to compensate for the different positions of the pivots **226a**, **226b** along the vertical direction **102c**.

(44) Referring to FIG. 2C, when transitioning from the collapsed configuration to the deployed configuration, the legs **112** may be first transitioned to the deployed configuration as shown with the tabletop frame **104** upside down and resting on a support surface. The braces **114** may be put in place to prevent the legs **112** from pivoting back toward the tabletop frame **104**. Alternatively, a detent or other fastener may be used to prevent the legs **112** from pivoting away from the deployed configuration. Referring to FIG. 2D, a user may then flip the table **100** such that the legs **112** are positioned between the tabletop frame **104** and the support surface. As shown, in both the partially deployed and collapsed configurations, the central segments **214** are positioned within an outer perimeter of the tabletop frame **104**.

(45) Referring to FIG. 2E, the user may then urge the central segments **214** of the collapsible sides **204a**, **204b** outwardly from one another, which results in the first and second end segments **116a**, **116b** also being forced outwardly from one another. Alternatively or additionally, the user may pull the first and second end segments **116a**, **116b** apart from one another, thereby causing the central segments **214** of the collapsible sides **204a**, **204b** to move outwardly from one another. The constrained movement imposed by the control arms **220a**, **220b** may facilitate the above-described

movement of the central segments **214** and the first and second end segments **116a**, **116b**.

Accordingly, in various embodiments, the control arms **220a**, **220b** ensure that deployment of the tabletop frame **104** proceeds in a controlled manner (e.g., with the central segments **214** remaining substantially parallel to one another and the end segments **116a**, **116b** remaining substantially parallel to one another), improving ease of use and enabling the tabletop frame **104** to be more reliably used in an intermediate configuration, as discussed below in further detail in conjunction with FIG. 2E.

(46) The user may urge the central segments **214** outwardly from one another until the deployed configuration of FIG. 2F is reached. As shown, in the deployed configuration, the central segments **214** form portions of the edges and outer perimeter of the tabletop frame **104**. In some embodiments, the tabletop frame **104** may also be used in the intermediate configuration shown in FIG. 2E. For example, in the intermediate configuration, N-1 (e.g., two) panels **106** may be rested on the tabletop frame **104**, whereas N (e.g., three) panels **106** will fit on the tabletop frame **104** in the deployed configuration (see, e.g., FIGS. 1B and 1C), where N is an integer greater than two. As shown in FIG. 2F, in the deployed configuration, the pivotable portions **206a**, **206b** are seated within the cavities **212** of the extensions **210a**, **210b** and within the cavity **218** of the central segment **214**, which resists flexing of the collapsible sides **204a**, **204b** in planes parallel to the vertical direction **102c**.

(47) Referring to FIG. 2G while still referring to FIG. 2F, the tabletop frame **104** may define a plurality of registration structures **228** that engage corresponding registration structures **230** formed on the panels **106**. The engagement of the registration structures **228**, **230** with one another both (a) resists inward movement of the central segments **214** toward one another and (b) resists sliding of the panels **106** relative to the tabletop frame **104**. In the illustrated embodiment, the registration structures **228** are embodied as openings in the upper surface **104a** of the tabletop frame **104** and the registration structures **230** are embodied as pins that are sized and positioned to insert within the openings when the tabletop frame **104** is in the deployed configuration. The pins may extend from a surface **106b** of the panels **106** opposite the planar surface **106a**. The reverse is also possible: pins secured to the tabletop frame **104** may engage openings defined by the panels **106**. In some embodiments, a portion of the panels **106**, such as two of the illustrated three panels **106**, may have registration structures **230** configured to engage the registration structures **228** when the tabletop frame **104** is in the intermediate configuration.

(48) As shown in FIG. 2G, some or all of the panels **106** may include one or more ribs **232** on the surface **106b** opposite the planar surface **106a**. The ribs **232** may be secured to the panels **106** or formed with the panels **106** by co-molding or some other process. The ribs **232** may serve to stiffen the panels **106** to prevent deflection during use. The ribs **232** may additionally facilitate aligning the panels **106** between the collapsible sides **204a**, **204b**. The ribs **232** may be oriented substantially (e.g., within 10 degrees of) parallel to the longitudinal direction **102b** when the panels **106** are placed on the tabletop frame **104**, though other configurations may also be used.

(49) Referring to FIG. 3A, the table **100** in the collapsed configuration may be sized to fit within a storage compartment **300** of a vehicle, such as the frunk or trunk of a vehicle **302**. The storage compartment **300** may include one or more walls **304** that are oriented substantially vertically (e.g., parallel to gravity with the vehicle **302** resting on a flat surface) or slightly slanted (e.g., within 20 degrees relative to vertical). The walls **304** may conform to a rectangle in planes perpendicular to vertical. The storage compartment **300** may further include one or more walls **306** that are non-rectangular in planes perpendicular to vertical, such as the illustrated walls **306** that are oriented diagonally (e.g., between 25 and 65 degrees from parallel or perpendicular to the walls **306**) or rounded corners between the walls **304**. The dimensions **108a**, **108b** of the table **100** in the collapsed configuration may be selected to fit within the largest rectangular shape that may be defined on a floor **308** of the storage compartment **300** extending across an area between the walls **304**, **306** of the storage compartment **300**.

(50) Referring to FIG. 3B, in other embodiments, the tabletop frame **104** and one or more of the panels **106** may include beveled corners **310** sized and shaped to conform to the non-rectangular walls **306**, thereby increasing the size of the table **100** that can fit within the storage compartment. Referring to FIG. 3C, in other embodiments and/or methods of use, the table **100** in the collapsed configuration is positioned within a cargo area **312** of a vehicle **314** located in front of a tailgate **316** and/or rear liftgate, or rear door of the vehicle **314** and behind the seats **318** of the vehicle **314**. The vehicle **314** may be embodied as a sport utility vehicle, hatchback, or other type of vehicle.

(51) In the methods of use shown in FIGS. 3A, 3B, and 3C, the table **100** is rested flat on the floor **308** or cargo area **312**. However, other methods of use are possible, such as with the table **100** oriented vertically (dimensions **108a** or **108b** oriented substantially (e.g., within 15 degrees of) parallel to vertical).

(52) Referring to FIG. 4, a method **400** for deploying a table, such as the table **100**, includes translating, at block **402**, a first central segment and a second central segment of the table, such as the central segments **214** of the collapsible sides **204a**, **204b**, outwardly from one another to cause one or more dimensions of the table to be modified, such as by modifying the dimension **108a** to the dimension **108c**, as described above. The table may include a plurality of joints configured to facilitate the translation of the first central segment and the second central segment, such as the pilotable portions **206a**, **206b** and the central segments **214** of the collapsible sides **204a**, **204b**. The method **400** further includes placing, at block **404**, one or more panels on the table to form a tabletop, such as the panels **106** described above.

(53) In one or more embodiments, movement of the first central segment and the second central segment modifies the table from having a first length to having a second length, such as from having the dimension **108a** to having the dimension **108c**. The first central segment and the second central segment may support the tabletop of the table. The first length may be an integer multiple of the second length, and a number of the one or more panels may be equal to the integer multiple. The first length may correspond to a collapsed configuration, and the second length may correspond to a deployed configuration. The first central segment and the second central segment may be substantially parallel to one another in both the collapsed configuration and the deployed configuration. In some embodiments, the first central segment and the second central segment are positioned within an outer perimeter of the table in the collapsed configuration and positioned at the outer perimeter of the table in the deployed configuration. For example, as shown in FIG. 2A, in a collapsed configuration, the central segments **214** are positioned within the perimeter of the tabletop frame **104** and, as shown in FIG. 2F, in the deployed configuration, the central segments **214** are positioned at the perimeter of the tabletop frame **104**.

(54) In some embodiments, the method **400** optionally includes translating a first end segment and a second end segment, such as the first and second end segments **116a**, **116b**, outwardly from one another. A first collapsible side including the first central segment may be secured to the first end segment and the second end segment, and a second collapsible side including the second central segment may be secured to the first end segment and the second end segment, the first collapsible side and the second collapsible side including the plurality of joints. The method **400** may further include securing a plurality of legs to the first end segment and the second end segment, such as the legs **112** as described above. In some embodiments, one or both of (a) the first and second collapsible sides and (b) the first and second end segments define one or more first registration features, and the one or more panels define one or more second registration features, such as the registration features **228**, **230**, as described above.

(55) Placing the one or more panels on the table may include engaging the one or more first registration features with the one or more second registration features such that the one or more panels resist inward movement of the first and second central segments. Securing the plurality of legs to the first end segment and the second end segment may include pivoting the plurality of legs outwardly from the first end segment and the second end segment.

(56) The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

(57) In the preceding, reference is made to embodiments presented in this disclosure. However, the scope of the present disclosure may exceed the specific described embodiments. Instead, any combination of the features and elements, whether related to different embodiments, is contemplated to implement and practice contemplated embodiments. Furthermore, although embodiments disclosed herein may achieve advantages over other possible solutions or over the prior art, the embodiments may achieve some advantages or no particular advantage. Thus, the aspects, features, embodiments and advantages discussed herein are merely illustrative.

(58) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. An apparatus, comprising: a table comprising: a first central segment; a first pivotable portion pivotally secured to an extension and pivotally secured to the first central segment; a first control arm pivotally secured to the extension and pivotally secured to the first central segment; a second central segment; and a plurality of joints configured to facilitate movement of the first central segment and the second central segment, wherein the movement of the first central segment and the second central segment causes one or more dimensions of the table to be modified.
2. The apparatus of claim 1, wherein the movement of the first central segment and the second central segment modifies the table from having a first length to having a second length, and wherein the first central segment and the second central segment are configured to support a tabletop.
3. The apparatus of claim 2, wherein: the second length is an integer multiple of the first length, and at the second length, the table is configured to support the tabletop comprising a plurality of panels, a number of the plurality of panels being equal to the integer multiple.
4. The apparatus of claim 2, wherein the first length corresponds to a collapsed configuration of the table, the second length corresponds to a deployed configuration of the table, and the first central segment and the second central segment remain substantially parallel to one another while the table is transitioned between the collapsed configuration and the deployed configuration.
5. The apparatus of claim 4, wherein the first central segment and the second central segment are: positioned within an outer perimeter of the table in the collapsed configuration, and positioned at the outer perimeter of the table in the deployed configuration.
6. The apparatus of claim 1, wherein the table further comprises: a first collapsible side comprising the first central segment, the first collapsible side secured to a first end segment and a second end segment; and a second collapsible side comprising the second central segment, the second collapsible side secured to the first end segment and the second end segment, wherein the first collapsible side and the second collapsible side comprise the plurality of joints.
7. The apparatus of claim 6, wherein: the first collapsible side further comprises: the first pivotable portion pivotally secured to the first end segment via the extension; and a second pivotable portion pivotally secured to the second end segment, wherein the first central segment is pivotally secured to the first pivotable portion and pivotally secured to the second pivotable portion; and the second collapsible side further comprises: a third pivotable portion pivotally secured to the first end segment; and a fourth pivotable portion pivotally secured to the second end segment, wherein the

second central segment is pivotally secured to the third pivotable portion and pivotally secured to the fourth pivotable portion.

8. The apparatus of claim 7, wherein: the first collapsible side further comprises: the first control arm pivotally mounted to the first end segment via the extension; and a second control arm pivotally mounted to the second end segment and pivotally mounted to the first central segment; and the second collapsible side further comprises: a third control arm pivotally mounted to the first end segment and pivotally mounted to the second central segment; and a fourth control arm pivotally mounted to the second end segment and pivotally mounted to the second central segment.

9. The apparatus of claim 8, wherein the first pivotable portion and the second pivotable portion are pivotally mounted to the first central segment between the first control arm and second control arm, and the third pivotable portion and the fourth pivotable portion are pivotally mounted to the second central segment between the third control arm and fourth control arm.

10. The apparatus of claim 6, wherein one or both of (a) the first and second collapsible sides and (b) the first and second end segments define a plurality of first registration features, the apparatus further comprising: one or more panels configured to rest on at least one of the first and second collapsible sides and the first and second end segments to form a tabletop, the one or more panels defining a plurality of second registration features configured to engage the plurality of first registration features.

11. The apparatus of claim 6, wherein the table comprises a plurality of legs secured to the first end segment and the second end segment, the plurality of legs being pivotable into (a) a collapsed configuration substantially parallel to the first end segment and the second end segment and (b) a deployed configuration substantially perpendicular to the first end segment and the second end segment.

12. The apparatus of claim 2, wherein the tabletop defines one or more features configured to mount an accessory comprising one or more of: a basin; a stove; and a storage structure.

13. A method for deploying a table, comprising: translating a first central segment and a second central segment of the table outwardly from one another to cause one or more dimensions of the table to be modified, wherein the table comprises a plurality of joints configured to facilitate the translation of the first central segment and the second central segment, wherein a first pivotable portion is pivotally secured to an extension and pivotally secured to the first central segment, and a first control arm is pivotally secured to the extension and pivotally secured to the first central segment; and placing one or more panels on the table to form a tabletop.

14. The method of claim 13, wherein movement of the first central segment and the second central segment modifies the table from having a first length to having a second length, and wherein the first central segment and the second central segment are configured to support the tabletop of the table.

15. The method of claim 14, wherein the first length is an integer multiple of the second length, and a number of the one or more panels is equal to the integer multiple.

16. The method of claim 14, wherein the first length corresponds to a collapsed configuration, the second length corresponds to a deployed configuration, and the first central segment and the second central segment remain substantially parallel to one another while the table is transitioned between the collapsed configuration and the deployed configuration.

17. The method of claim 16, wherein the first central segment and the second central segment are: positioned within an outer perimeter of the table in the collapsed configuration, and positioned at the outer perimeter of the table in the deployed configuration.

18. The method of claim 13, further comprising: translating a first end segment and a second end segment outwardly from one another, wherein a first collapsible side comprising the first central segment is secured to the first end segment and the second end segment, and a second collapsible side comprising the second central segment is secured to the first end segment and the second end segment, the first collapsible side and the second collapsible side comprising the plurality of joints;

and securing a plurality of legs to the first end segment and the second end segment.

19. The method of claim 18, wherein one or both of (a) the first and second collapsible sides and (b) the first and second end segments define one or more first registration features, and the one or more panels define one or more second registration features, and wherein placing the one or more panels on the table comprises engaging the one or more first registration features with the one or more second registration features such that the one or more panels resist inward movement of the first and second central segments.

20. The method of claim 18, wherein securing the plurality of legs to the first end segment and the second end segment comprises pivoting the plurality of legs outwardly from the first end segment and the second end segment.
